Study notes: Module IV: dielectric and magnetic properties of solids

Puri and Babbar, *Solid State Physics and Electronics*, chapters 8 and 9 (printed pages 228–279). Text extracted by Codex from the page images and checked against them. Figures are redrawn for these notes, not copied from the book; the book figure number is in each caption. Chapter questions and problems are not included.

Section 1: magnetic terminology and types of magnetism (8.1–8.2)

8.1–8.2 Magnetic terminology and types of magnetism (student notes)

Source: book-sources/bs-6-magnetism.html, Puri and Babbar pp. 228–230. Equations 8.1 to 8.8 are cited by their book numbers.

Before you start

- A magnetic dipole (a bar magnet or a current loop) and the field it makes (Plus Two recall below).
- Electrons as moving charges: an orbiting electron is a small current.
- Magnetic field $\mathbf{B}$ in tesla, and $\mu_0 = 4\pi \times 10^{-7}$ H/m.

> **PLUS TWO RECALL: MAGNETIC MOMENT AND MAGNETIC MATERIALS**
>
> **Recall.** A current loop of area A carrying current I has magnetic moment $m = IA$. Materials are classed as diamagnetic, paramagnetic or ferromagnetic by how they behave in a field.
>
> **Re-derive.**
>
> 1. A magnetised solid is a collection of small moments. Their sum per unit volume is the magnetisation $\mathbf{M}$.
> 2. In a weak field $\mathbf{M}$ is proportional to $\mathbf{H}$, so the ratio $\chi = M/H$ is a constant of the material.
> 3. Materials with χ negative, small and positive, or large and positive are the three classes that Plus Two names.
>
> **Saved.** Eq. (8.1) for χ. The magnetic moment of a loop is the basis of §8.3 and §8.4.

1. Magnetisation and susceptibility (§8.1)

A solid placed in a magnetic field becomes magnetised. The magnetic moment per unit volume is the magnetisation $\mathbf{M}$. The susceptibility χ is the magnetisation produced per unit applied field:

$$\chi = M/H \tag{8.1}$$

Here $\mathbf{H}$ is the magnetic field intensity. In general $\mathbf{M}$ and $\mathbf{H}$ can point in different directions, so χ is a tensor. In an isotropic medium they point the same way and χ is a scalar. In SI units M and H are in A/m, so χ is dimensionless. If M is per gram molecule, the value is the molar susceptibility χ_m.

The magnetic flux density inside the material is given by

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) \tag{8.2}$$

where $\mu_0 = 4\pi \times 10^{-7}$ H/m is the permeability of free space. Using (8.1) in (8.2):

$$\mathbf{B} = \mu_0(1 + \chi)\mathbf{H} \tag{8.3}$$

This can also be written with the absolute permeability μ:

$$\mathbf{B} = \mu\mathbf{H} \tag{8.4}$$

It is more convenient to use the relative permeability μ_r, defined by

$$\mu = \mu_0\mu_r \tag{8.5}$$

so that (8.4) becomes

$$\mathbf{B} = \mu_0\mu_r\mathbf{H} \tag{8.6}$$

Comparing (8.3) and (8.6) gives the link between permeability and susceptibility:

$$\mu_r = 1 + \chi \tag{8.7}$$

In free space $M = 0$, $\chi = 0$, $\mu_r = 1$, and so $\mathbf{B} = \mu_0\mathbf{H}$ (8.8).

$$\mathbf{B} = \mu_0\mathbf{H} \tag{8.8}$$

2. Where magnetism comes from and the five types (§8.2)

Magnetism in solids comes from the orbital and spin motion of electrons, and a much smaller part from the nuclear spins. The main contribution is the spin of unpaired valence electrons, which gives a permanent electronic moment. Whether a solid is magnetic, and how, depends on how many unpaired electrons its atoms have and on how neighbouring moments line up.

The book names five types:

- **Diamagnetism**: a very weak effect in solids with no permanent moments. The moment is induced by the orbital motion of electrons and always opposes the field.
- **Paramagnetism**: a weak effect from permanent moments. The moments line up with the field, so the effect is positive.
- **Ferromagnetism**: a very strong effect. Adjacent moments line up in the same direction, even with no field.
- **Antiferromagnetism**: adjacent moments are equal and opposite, so they cancel completely.
- **Ferrimagnetism**: like antiferromagnetism, but the adjacent moments are unequal, so they do not cancel.

Above a critical temperature the differences between these types disappear and every material is paramagnetic. Sections 8.3 to 8.7 take each type in turn.

Section 2: diamagnetism, classical and quantum (8.3, Example 8.1)

8.3 Diamagnetism (student notes)

Source: book-sources/bs-6-magnetism.html, Puri and Babbar pp. 230–235 (§8.3, §8.3.1 classical, §8.3.2 quantum, Example 8.1). Equations 8.9 to 8.28 are cited by their book numbers. The quantum theory (§8.3.2) is background to item 18; it is kept in full because the book gives its equations.

Before you start

- Lenz's law: an induced current opposes the change that produces it (Plus Two recall below).
- Coulomb's law for the nucleus pulling an orbiting electron, and the centripetal force $m\omega^2\rho$.
- The magnetic moment of a current loop, $\mu = IA$, and the vector potential $\mathbf{B} = \nabla \times \mathbf{A}$ (background, §8.3.2).

> **PLUS TWO RECALL: LENZ'S LAW AND OPPOSING FLUX**
>
> **Recall.** When the flux through a circuit changes, the induced current opposes the change.
>
> **Re-derive.**
>
> 1. An electron in a circular orbit is a current loop, so it carries a magnetic moment.
> 2. Switching on a field changes the flux through that loop.
> 3. By Lenz's law the induced current, and so the induced moment, opposes the field.
>
> **Saved.** The induced moment is antiparallel to B, giving a negative susceptibility (Eq. 8.16, 8.20).

1. The classical (Langevin) picture (§8.3.1)

Take one electron in a circular orbit of radius ρ and angular frequency ω_0 about a nucleus of charge Ze. The electrostatic pull supplies the centripetal force:

$$m\omega_0^2\rho = \frac{Ze^2}{4\pi\varepsilon_0\rho^2} \tag{8.9}$$

This gives

$$\omega_0 = \sqrt{\frac{Ze^2}{4\pi\varepsilon_0 m\rho^3}} \tag{8.10}$$

Now apply a field $\mathbf{B}$ perpendicular to the orbit. The Lorentz force on the electron is

$$\mathbf{F} = e(\mathbf{v}\times\mathbf{B}) \tag{8.11}$$

whose magnitude, for a circular orbit, is

$$F = e\rho\omega B \tag{8.12}$$

With the field, the frequency becomes ω. The force balance is

$$\begin{aligned} m\omega^2\rho &= \frac{Ze^2}{4\pi\varepsilon_0\rho^2} - e\rho\omega B \\ &= m\omega_0^2\rho - e\rho\omega B \end{aligned} \tag{8.13}$$

The minus sign comes from the negative charge. This leads to $\omega^2 + (eB/m)\omega - \omega_0^2 = 0$, whose root for $eB/2m \ll \omega_0$ is

$$\omega \cong \pm\omega_0 - \frac{eB}{2m} \tag{8.14}$$

The frequency changes by $eB/2m$. Electrons whose orbital moment is parallel to the field slow down; antiparallel ones speed up. This is Larmor's theorem. The change in frequency gives an extra current:

$$\begin{aligned} I &= \text{charge} \times \text{revolutions per unit time} \\ &= (-e)\left(\frac{1}{2\pi}\frac{eB}{2m}\right) \\ &= -\frac{e^2 B}{4\pi m} \end{aligned} \tag{8.15}$$

The moment of this loop is current times area:

$$\mu_e = -\frac{e^2 B}{4m}\rho^2 \tag{8.16}$$

The minus sign shows the induced moment is opposite to the field. For Z electrons with randomly oriented orbits the total is

$$\mu_a = -\frac{Ze^2 B}{4m}\langle\rho^2\rangle \tag{8.17}$$

For a spherically symmetric charge distribution $\langle x^2\rangle = \langle y^2\rangle = \langle z^2\rangle$, so the mean square radius relative to the nucleus is

$$\langle r^2\rangle = \frac{3}{2}\langle\rho^2\rangle \tag{8.18}$$

Then (8.17) becomes

$$\begin{aligned} \mu_a &= -\frac{Ze^2 B}{6m}\langle r^2\rangle \\ &= -\frac{\mu_0 Ze^2 H}{6m}\langle r^2\rangle \end{aligned} \tag{8.19}$$

For N atoms per unit volume the susceptibility is

$$\chi_{dia} = \frac{M}{H} = -\frac{N\mu_0 Ze^2}{6m}\langle r^2\rangle \tag{8.20}$$

This is the classical Langevin result. It is independent of temperature.

2. The quantum treatment (§8.3.2, background)

In quantum mechanics the Hamiltonian of an atomic electron changes by a small term $\mathcal{H}'$ in a field:

$$\begin{aligned} \mathcal{H}' = &-\frac{ie\hbar}{2m}(\nabla\cdot\mathbf{A} + 2\mathbf{A}\cdot\nabla) \\ &+ \frac{e^2}{2m}A^2 \end{aligned} \tag{8.21}$$

where $\mathbf{B} = \nabla\times\mathbf{A}$. For a uniform field,

$$\mathbf{A} = \frac{1}{2}\mathbf{B}\times\mathbf{r} \tag{8.22}$$

Taking $\mathbf{B}$ along z:

$$A_x = -\frac{1}{2}yB, \quad A_y = \frac{1}{2}xB, \quad A_z = 0 \tag{8.23}$$

Also $\nabla\cdot\mathbf{A} = 0$, and $\mathbf{A}\cdot\nabla = \frac{1}{2}B(x\partial_y - y\partial_x)$. So (8.21) becomes

$$\begin{aligned} \mathcal{H}' = &-\frac{ie\hbar B}{2m}\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right) \\ &+ \frac{e^2B^2}{8m}(x^2+y^2) \end{aligned} \tag{8.24}$$

The first term is proportional to L_z, the z-component of the angular momentum operator:

$$\begin{aligned} \mathbf{L} &= -i\hbar\,\mathbf{r}\times\nabla \\ &= -i\hbar\left[\hat{\mathbf{i}}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)\right. \\ &\quad \left. + \hat{\mathbf{j}}\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right) + \hat{\mathbf{k}}\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right)\right] \end{aligned} \tag{8.25}$$

For a unit volume of N atoms with Z electrons each, the second term gives

$$\frac{Ne^2B^2}{8m}\sum_{i=1}^{Z}(x_i^2 + y_i^2) = \frac{NZe^2B^2}{8m}\langle\rho^2\rangle \tag{8.26}$$

This energy is of the second power in B, as a diamagnetic energy must be. Comparing it with $-\frac{1}{2\mu_0}\chi_{dia}B^2$ gives

$$\chi_{dia} = -\frac{N\mu_0 Ze^2}{4m}\langle\rho^2\rangle \tag{8.27}$$

Using (8.18):

$$\chi_{dia} = -\frac{N\mu_0 Ze^2}{6m}\langle r^2\rangle \tag{8.28}$$

This is the same as the classical result. In quantum mechanics $\langle r^2 \rangle$ is found from the wave function, not from orbits. Agreement with experiment is good for light elements and rare gases. Helium has $\chi \approx -1.9 \times 10^{-6}$, which shows diamagnetism is very weak.

3. Worked Example 8.1: diamagnetism of atomic hydrogen

For the ground state of hydrogen, $\psi(r) = (\pi a_0^3)^{-1/2} \exp(-r/a_0)$ with $a_0 = 0.46$ Å. The mean square distance is $\langle r^2 \rangle = \int \psi^* r^2 \psi \, d\mathbf{r} = 3a_0^2$. Equation (8.28) with $Z = 1$ gives $\chi_{dia} = -(N\mu_0 e^2/2m)a_0^2$. With $N = 2.69 \times 10^{28}$ m^{-3} and $a_0 = 0.46 \times 10^{-10}$ m, the book gives $\chi_{dia} = -1.01 \times 10^{-6}$ (SI).

Section 3: paramagnetism, Langevin and quantum theories (8.4, Examples 8.2 and 8.3)

8.4 Paramagnetism (student notes)

Source: book-sources/bs-6-magnetism.html, Puri and Babbar pp. 235–242 (§8.4, §8.4.1 classical, §8.4.2 quantum; Examples 8.2 and 8.3). Equations 8.29 to 8.51 and 8.83 are cited by their book numbers. Figures 8.1 and 8.2 are redrawn.

Before you start

- Potential energy of a magnetic dipole in a field, $U = -\boldsymbol{\mu} \cdot \mathbf{B}$ (Plus Two recall below).
- The Maxwell–Boltzmann factor $\exp(-E/kT)$ (given in the pre files as Beyond).
- Integration by the substitution $y = \cos\theta$ (Module III pre-math, M5).

PLUS TWO RECALL: A MAGNETIC DIPOLE IN A FIELD

Recall. A magnetic dipole of moment $\boldsymbol{\mu}$ in a uniform field $\mathbf{B}$ has potential energy $U = -\boldsymbol{\mu} \cdot \mathbf{B}$, and it tends to align with the field.

Re-derive.

1. At an angle θ to the field, $U = -\mu B \cos\theta$.
2. The lowest energy is at $\theta = 0$, where the dipole lies along the field.
3. Thermal agitation spreads the dipoles over all angles, so the net moment is only partly aligned.

Saved. The energy $U = -\mu B \cos\theta$ is used in §8.4.1 (Fig. 8.1).

1. Paramagnetic materials

Paramagnetism is found in atoms, ions and molecules that carry permanent moments. With no field the moments point at random, so there is no net magnetisation. A field lines them up partly, giving a small positive susceptibility that depends on temperature. The moments come from electron spin, electron orbital motion and nuclear spin.

Paramagnetism occurs in metals, in atoms and molecules with an odd number of electrons (free sodium atoms, nitric oxide), in a few molecules with an even number (oxygen), and in free atoms and ions with a partly filled inner shell (rare earths, Mn^{2+}).

2. Langevin's classical theory (§8.4.1)

Langevin considered N atoms per unit volume, each with a permanent moment μ, and neglected their mutual interaction. A dipole at angle θ to the field (Fig. 8.1) has energy $E = -\mu B\cos\theta$. By the Maxwell–Boltzmann distribution, the number of dipoles in the solid angle $d\omega$ is proportional to $\exp(\mu B\cos\theta/kT)$.

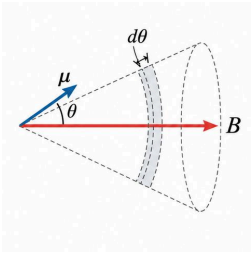


Fig. 8.1 (book): magnetic dipole with moment μ oriented at an angle θ to the applied magnetic field.

$$dn \propto \exp\left(\frac{\mu B\cos\theta}{kT}\right)d\omega = k\exp\left(\frac{\mu B\cos\theta}{kT}\right)2\pi\sin\theta\,d\theta \tag{8.29}$$

Each dipole contributes $\mu\cos\theta$ along the field. The components across the field cancel. So the magnetisation is the average of $\mu\cos\theta$, times N:

$$\begin{aligned} M &= N\mu\langle\cos\theta\rangle \\ &= N\frac{\int_0^{\pi}\mu\cos\theta\,dn}{\int_0^{\pi}dn} \\ &= N\frac{\mu\int_0^{\pi}\cos\theta\sin\theta\exp\left(\frac{\mu B\cos\theta}{kT}\right)d\theta}{\int_0^{\pi}\sin\theta\exp\left(\frac{\mu B\cos\theta}{kT}\right)d\theta} \end{aligned} \tag{8.30}$$

Write $x = \mu B/kT$ and $y = \cos\theta$, so $-\sin\theta\,d\theta = dy$. Then the integrals become exponentials, and

$$\begin{aligned} M &= N\mu\frac{\int_1^{-1} y e^{xy}\,dy}{\int_1^{-1} e^{xy}\,dy} \\ &= N\mu\left[\frac{e^{x}+e^{-x}}{e^{x}-e^{-x}} - \frac{1}{x}\right] \\ &= N\mu\left[\coth x - \frac{1}{x}\right] \\ &= N\mu\,L(x) \end{aligned} \tag{8.31}$$

$L(x)$ is the Langevin function (Fig. 8.2).

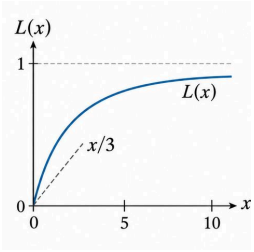


Fig. 8.2 (book): variation of L(x) with x.

For $x \ll 1$, which means normal fields and ordinary temperatures, $L(x)$ is close to its tangent at the origin, $x/3$:

$$L(x)\big|_{x=0} \cong \frac{x}{3} = \frac{\mu B}{3kT} \tag{8.32}$$

So $M \cong N\mu^2 B/3kT$, and the susceptibility is

$$\chi_{para} = \frac{\mu_0 M}{B} = \frac{\mu_0 N\mu^2}{3kT} \tag{8.33}$$

This is inversely proportional to temperature:

$$\chi_{para} = \frac{C}{T} \tag{8.34}$$

with $C = \mu_0 N\mu^2/3k$, the Curie constant. The Curie law holds when $\mu B \ll kT$. For large x (high fields, low temperatures) $L(x) \to 1$, and the magnetisation saturates:

$$M = N\mu = M_s \tag{8.35}$$

3. Quantum theory of paramagnetism (§8.4.2)

In quantum theory the moments are quantised. The moment is related to the total angular momentum $\mathbf{J}$ by

$$\boldsymbol{\mu} = -g\mu_B\mathbf{J} \tag{8.36}$$

Here $\mu_B = e\hbar/2m$ is the Bohr magneton, and g is the Landé g-factor:

$$g = 1 + \frac{J(J+1)+S(S+1)-L(L+1)}{2J(J+1)} \tag{8.37}$$

g is 2 for pure spin and 1 for pure orbital motion. The component along the field is

$$\mu_z = -g\mu_B m_J \tag{8.38}$$

where $m_J = -J,\ldots,+J$ takes $(2J+1)$ values. The energy of each state is

$$E = m_J g\mu_B B \tag{8.39}$$

The magnetisation is the Boltzmann-weighted average over these states:

$$M = N\sum_{m_J=-J}^{m_J=+J}\frac{-m_J g\mu_B e^{-m_J g\mu_B B/kT}}{e^{-m_J g\mu_B B/kT}} \tag{8.40}$$

Two cases follow.

(i) Normal fields and ordinary temperatures. Here $m_J g \mu_B B / kT \ll 1$, so the exponentials are expanded. Using $\sum m_J = 0$ and $\sum m_J^2 = J(J+1)(2J+1)/3$:

$$M = N \frac{g^2 \mu_B^2 B}{3kT} J(J+1) \tag{8.41}$$

so

$$\chi_{para} = \frac{\mu_0 M}{B} = \frac{\mu_0 N \mu_B^2}{3kT} g^2 J(J+1) \tag{8.42}$$

$$\chi_{para} = \frac{\mu_0 N p_{eff}^2 \mu_B^2}{3kT} \tag{8.43}$$

where the effective number of Bohr magnetons is

$$p_{eff} = g\sqrt{J(J+1)} \tag{8.44}$$

(8.43) has the classical form (8.33) with

$$\mu^2 = p_{eff}^2 \mu_B^2 \tag{8.45}$$

so the Curie law follows as before.

(ii) Low temperatures and strong fields. Now $gJ\mu_B B/kT$ is not small, and the magnetisation is

$$M = N g J \mu_B B_J(x) \tag{8.46}$$

with $x = gJ\mu_B B/kT$ and the Brillouin function

$$B_J(x) = \frac{2J+1}{2J} \coth\!\left(\frac{2J+1}{2J}x\right) - \frac{1}{2J}\coth\!\left(\frac{x}{2J}\right) \tag{8.47}$$

For $x \ll 1$, $\coth x \cong 1/x + x/3$, which gives $B_J(x) = x(J+1)/3J$. Then the susceptibility is

$$\chi_{para} = \frac{\mu_0 M}{B} = \frac{\mu_0 N J(J+1) g^2 \mu_B^2}{3kT} = \frac{\mu_0 N p_{eff}^2 \mu_B^2}{3kT} \tag{8.48}$$

which is (8.43) again. For $x \gg 1$, $B_J \cong 1$ and the saturation value is

$$M = N g J \mu_B \tag{8.49}$$

For $J \to \infty$ the Brillouin function becomes $L(x)$, so the classical result is the limit of the quantum one. For other J the two differ.

At room temperature (8.48) gives a susceptibility of order 10^{-7}, and it grows by more than a hundredfold at 1 K. Hund's rules give J. Equation (8.48) predicts rare-earth ions well, but not ions of the iron group, because the crystal field can quench the orbital moment. Then

$$E = 2 m_s \mu_B B \tag{8.50}$$

with $g = 2$, and

$$\chi = \frac{\mu_0 N}{3kT}\left(\frac{e\hbar}{m}\right)^2 S(S+1) \tag{8.51}$$

For Mn^{3+} with $S = 2$ this agrees with experiment, where (8.48) predicts zero.

4. Worked Examples 8.2 and 8.3

Example 8.2. A paramagnetic gas with $L = 0$, $S = \tfrac{1}{2}$ has $J = \tfrac{1}{2}$, $m_J = \pm\tfrac{1}{2}$ and $g = 2$. The two levels have populations $N_{\pm 1/2}$ with $N_{1/2} + N_{-1/2} = N$. The magnetisation is

$$M = g\mu_B J\left(N_{-1/2} - N_{1/2}\right) = g\mu_B J N \tanh\!\left(\frac{g\mu_B B}{2kT}\right) \tag{8.83}$$

For $N = 10^{28}$ m^{-3} and $B = 2.5$ T, the book gives $N_{1/2} = 5 \times 10^{27}\,\exp(1.68/T)/\cosh(1.68/T)$ and $M = 9.27 \times 10^{4} \tanh(1.68/T)$. At 300 K this gives $N_{1/2} = 4.97 \times 10^{27}$ m^{-3}, $N_{-1/2} = 5.03 \times 10^{27}$ m^{-3} and $M = 519.1$ A/m. At 5 K, $N_{1/2} = 3.38 \times 10^{27}$, $N_{-1/2} = 6.62 \times 10^{27}$ m^{-3} and $M = 3.0 \times 10^{4}$ A/m.

Example 8.3. Dy^{3+} ($4f^9$) has $S = 5/2$, $L = 5$ and $J = 15/2$ by Hund's rules. Then $g = 1.33$. For one kilogram-mole of salt, $N = 6.023 \times 10^{26}$, and the book gives $\chi = \frac{\mu_0 N \mu_B^2 g^2 J(J+1)}{3kT} = 5.9 \times 10^{-4}$ at 300 K.

Section 4: ferromagnetism: Weiss theory, exchange, domains and hysteresis (8.5, Example 8.4)

8.5 Ferromagnetism: Weiss theory, exchange, domains and hysteresis (student notes)

Source: book-sources/bs-6-magnetism.html, Puri and Babbar pp. 242–251 (§8.5, §8.5.1 to §8.5.3; Example 8.4). Equations 8.52 to 8.75 and 8.84 are cited by their book numbers. Figures 8.3 to 8.7 are redrawn.

Before you start

- The Brillouin function $B_J(x)$ and the saturated magnetisation $M_s = NgJ\mu_B$ from §8.4 (Eqs. 8.46, 8.47, 8.49).
- Spin and the Pauli exclusion principle (Plus Two recall below).
- The Curie–Weiss idea of a molecular field (Beyond the Plus Two base; §8.5.1 is the first time it appears).

> **PLUS TWO RECALL: MAGNETIC MATERIALS AND HYSTERESIS**
>
> **Recall.** A soft magnetic material is easily magnetised and demagnetised; a hard one keeps its magnetisation. Their behaviour is shown by a loop in B against H.
>
> **Re-derive.**
>
> 1. Increase H from zero and the magnetisation rises to a saturation value.
> 2. Reduce H to zero: some magnetisation remains (remanence M_r).
> 3. Reverse H until the magnetisation is zero (coercivity H_c). The closed curve is the hysteresis loop.
>
> **Saved.** Hysteresis and the loop are developed in §8.5.3 (Fig. 8.6).

1. The Weiss theory (§8.5.1)

A ferromagnetic material has a spontaneous magnetisation below its Curie temperature T_C, with no field applied. Above T_C it is paramagnetic. Its susceptibility is large and positive, and depends on the field. Iron, nickel, cobalt, gadolinium and dysprosium are examples.

Weiss made two hypotheses. First, the specimen is divided into domains, each spontaneously magnetised. Second, each domain has an exchange field B_E that aligns the atomic dipoles, and this field is proportional to the magnetisation:

$$B_E = \lambda M \tag{8.52}$$

λ is the Weiss-field constant and does not depend on temperature. The field is strong: for iron $B_E \approx 1000$ T. The effective field on an atom is

$$B_{eff} = B + B_E = B + \lambda M \tag{8.53}$$

Weiss first used the classical Langevin theory. The quantum form is used here. The magnetisation is

$$M = NgJ\mu_B B_J(x) \tag{8.54}$$

with the Brillouin function (8.55), which is (8.47) with x defined by

$$B_J(x) = \frac{2J+1}{2J}\coth\left(\frac{2J+1}{2J}x\right) - \frac{1}{2J}\coth\left(\frac{x}{2J}\right) \tag{8.55}$$

$$x = \frac{gJ\mu_B B_{eff}}{kT} = \frac{gJ\mu_B}{kT}(B + \lambda M) \tag{8.56}$$

2. Spontaneous magnetisation

With no applied field, $B = 0$ and

$$x = \frac{gJ\mu_B \lambda M}{kT} \tag{8.57}$$

$$M(T) = \frac{xkT}{\lambda g J \mu_B} \tag{8.58}$$

As $T \to 0$, $B_J \to 1$, and the magnetisation becomes the saturation value

$$M_s(0) = NgJ\mu_B \tag{8.59}$$

Combining (8.58) and (8.59), and then with (8.54) written as a ratio:

$$\frac{M(T)}{M_s(0)} = \frac{xkT}{\lambda N g^2 J^2 \mu_B^2} \tag{8.60}$$

$$\frac{M(T)}{M_s(0)} = B_J(x) \tag{8.61}$$

The two equations are solved together graphically (Fig. 8.3). Equation (8.60) is a straight line through the origin whose slope is proportional to T. Its intersection with the Brillouin curve gives $M_s(T)$. For $T < T_C$ there is a non-zero intersection, so there is spontaneous magnetisation. At T_C the line is tangent to the curve at the origin. Above T_C there is no such intersection.

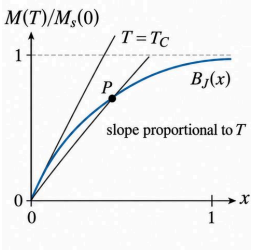


Fig. 8.3 (book): graphical solution of the simultaneous equations (8.60) and (8.61); the point of intersection P gives the spontaneous magnetisation $M_S(T)$.

The spontaneous magnetisation falls with temperature and vanishes at T_C (Fig. 8.4). This makes the transition a second-order phase transition, with M_s as the order parameter.

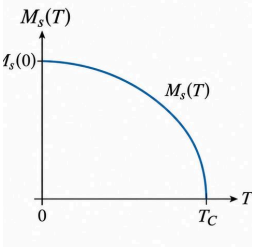


Fig. 8.4 (book): spontaneous magnetisation against temperature for $T < T_C$.

3. Above the Curie temperature

For $T > T_C$ the magnetisation is produced only by an applied field, and $x \ll 1$, so $B_J(x) \cong \frac{J+1}{3J}x$. Then (8.54) becomes $M = Ng\mu_B(J+1)x/3$, and with (8.56):

$$x = \frac{gJ\mu_B(B+\lambda M)}{kT} \tag{8.62}$$

$$M = \frac{Ng^2\mu_B^2 J(J+1)}{3kT}(B+\lambda M) \tag{8.63}$$

This gives the Curie–Weiss law:

$$\chi = \frac{\mu_0 M}{B} = \frac{\mu_0 T_C/\lambda}{T-T_C} = \frac{C}{T-T_C} \tag{8.64}$$

with

$$C = \frac{\mu_0 T_C}{\lambda} \tag{8.65}$$

$$T_C = \frac{\lambda N g^2 \mu_B^2 J(J+1)}{3k} \tag{8.66}$$

The law describes the paramagnetic region well, a few degrees above T_C. The Curie temperature from the spontaneous-magnetisation theory differs from the measured one by a few degrees.

4. The origin of the Weiss field: exchange (§8.5.2)

A dipole–dipole field would be of order 10^3 G, far too small. For iron the Weiss field is of order 10^7 G. Heisenberg proposed in 1928 that it comes from exchange interaction. This is electrostatic in origin, arising from the overlap of wave functions with the Pauli principle fixing the spin orientation. The Heisenberg exchange energy of two atoms is

$$U_{ij} = -2J_e \mathbf{S}_i \cdot \mathbf{S}_j \tag{8.67}$$

J_e is the exchange integral. It is positive at large separations and negative at small ones. A positive J_e favours parallel spins, which is why it gives magnetism. For iron-group atoms J_e is positive, and Bethe and Slater showed that $J_e > 0$ when r_{ij}/r_{3d} is greater than about 3 (Fig. 8.5). This favours Fe, Co, Ni and Gd, and not Mn or Cr.

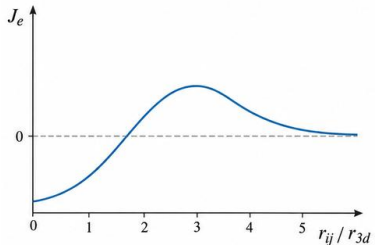


Fig. 8.5 (book): exchange integral J_e against the ratio of interatomic separation to the radius of the 3d orbit.

Stoner's method takes the time average of the neighbouring spins over z nearest neighbours:

$$U_i = -2J_e \sum_j \mathbf{S}_i \cdot \mathbf{S}_j \tag{8.68}$$

$$U_i = -2zJ_e\left(S_{xi}\langle S_{xj}\rangle + S_{yi}\langle S_{yj}\rangle + S_{zi}\langle S_{zj}\rangle\right) \tag{8.69}$$

For magnetisation M along z, $\langle S_{xj}\rangle = \langle S_{yj}\rangle = 0$ and

$$\langle S_{zj}\rangle = \frac{M}{g\mu_B N} \tag{8.70}$$

So

$$U_i = -\frac{2zJ_e S_{zi} M}{gN\mu_B} \tag{8.71}$$

$$U_i = -g S_{zi} \mu_B B_E \tag{8.72}$$

with

$$B_E = \frac{2zJ_e}{Ng^2\mu_B^2} M \tag{8.73}$$

The factor is the Weiss constant:

$$\lambda = \frac{2zJ_e}{Ng^2\mu_B^2} \tag{8.74}$$

and (8.73) reduces to $B_E = \lambda M$, which is (8.52).

5. Domains and hysteresis (§8.5.3)

A ferromagnetic crystal is divided into domains, each spontaneously magnetised. Their directions are random, so the crystal as a whole has no net magnetisation. An applied field makes the favourable domains grow at the expense of the others (Fig. 8.7b), and then rotates their magnetisation towards the field (Fig. 8.7c).

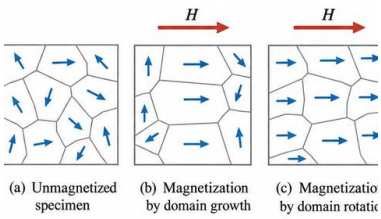


Fig. 8.7 (book): the two fundamental processes of magnetisation: (a) unmagnetised specimen; (b) magnetisation by domain growth; (c) magnetisation by domain rotation.

Neel explained the domain structure by minimising total energy: magnetic field energy, anisotropy energy, domain-wall energy and magnetostrictive energy. Subdividing a domain reduces the field energy, but each new wall costs energy, so a finite domain structure is the minimum.

Hysteresis is shown by all ferromagnets. For $T < T_C$, the two solutions of (8.61) and

$$M = \left(\frac{kT}{\mu_B g \lambda J}\right) x - \frac{1}{\lambda} B \tag{8.75}$$

trace the boundary of the hysteresis loop (Fig. 8.6). Starting from zero, the initial curve OA is reversible; AB is a large irreversible growth of domains; BC is slow rotation to saturation M_s. On reducing the field there is a remanence M_r. The field that brings M to zero is the coercive field H_c. The loop CEFKC is closed, and a B–H loop of similar shape is drawn. The loop shows the quality of a magnetic material and so the choice for an application.

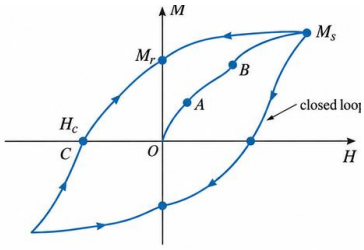


Fig. 8.6 (book): typical hysteresis curve for a ferromagnetic material, indicating the predominant magnetisation processes in different regions.

6. Worked Example 8.4: intrinsic flux density and the field ratio

For $J = 3/2$, $g = 2$ and $T_C = 125$ K: at 0 K the intrinsic flux density is $B_E = \lambda M_s(0)$. From (8.66), $B_E = 3kT_C/[g\mu_B(J+1)] = 111.6$ T. Above T_C the magnetisation follows (8.84):

$$M = \frac{Ng^2\mu_B^2 J(J+1) B}{3k(T - T_C)} \tag{8.84}$$

so $M/M_s(0) = g\mu_B B(J+1)/[3k(T - T_C)]$. For $B = 1$ mT and $T = 300$ K this gives $M/M_s(0) = 6.4 \times 10^{-6}$.

8.6–8.7 Antiferromagnetism and ferrimagnetism (student notes)

Source: book-sources/bs-6-magnetism.html, Puri and Babbar pp. 251–255 (§8.6 and §8.7). Equations 8.76 to 8.82 are cited by their book numbers. Figures 8.8 to 8.12 are redrawn. Fig. 8.9 is printed in the book with no caption; its caption says so. Item 19 is qualitative: the molecular-field algebra below is the book's, given for the picture, not for examination.

Before you start

- Curie–Weiss law $\chi = C/(T - T_C)$ (Eq. 8.64) and the paramagnetic law (8.33), from §8.4 and §8.5.
- Exchange interaction and its sign (§8.5.2): negative exchange favours antiparallel spins.

The Plus Two base does not cover these types; the pre file marks them Beyond. Antiparallel alignment (pre item P14) is the qualitative idea.

1. Antiferromagnetism (§8.6)

In an antiferromagnet the spins of neighbouring atoms are antiparallel (Fig. 8.8), or the exchange integral is negative. The crystal consists of two interpenetrating sublattices, A and B, magnetised in opposite directions. With no field the moments cancel and there is no magnetisation. MnO was the first crystal in which this was observed.

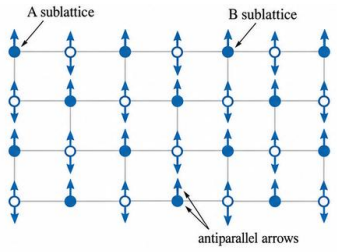


Fig. 8.8 (book): neighbouring antiparallel spin moments of the A and B sublattices producing antiferromagnetism.

A small field gives a small magnetisation in its direction, which grows with temperature up to the Néel temperature T_N. Above T_N the material is paramagnetic. The susceptibility has a weak cusp at T_N, not the infinity of a ferromagnet (Fig. 8.9).

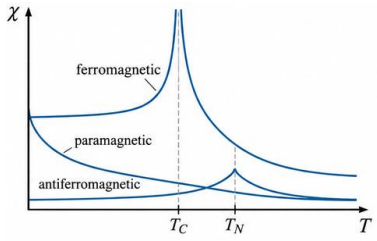


Fig. 8.9 (book): the book prints the number only; no caption is printed.

2. Néel's molecular-field picture

Each sublattice atom has nearest-neighbour negative AB interaction, and a weaker AA or BB interaction. With Weiss field constants α (for AA and BB) and β (for AB), the internal fields at sites A and B are

$$\begin{aligned} B_{mA} &= -\alpha M_A - \beta M_B \\ B_{mB} &= -\beta M_A - \alpha M_B \end{aligned} \tag{8.76}$$

With an external field B, the effective fields are

$$\begin{aligned} B_A &= B - \alpha M_A - \beta M_B \\ B_B &= B - \beta M_A - \alpha M_B \end{aligned} \tag{8.77}$$

Above T_N. Each sublattice is paramagnetic, with the paramagnetic law (8.33) applied to its field:

$$\begin{aligned} M_A &= \frac{N\mu^2}{3kT} B_A \\ M_B &= \frac{N\mu^2}{3kT} B_B \end{aligned} \tag{8.78}$$

where $\mu^2 = \mu_B^2 g^2 J(J+1)$. The two sublattices have equal densities, $N_A = N_B = N$. Adding them and using (8.77):

$$\begin{aligned} M &= M_A + M_B \\ &= \frac{N\mu^2}{3kT}(B_A + B_B) \\ &= \frac{N\mu^2}{3kT}[2B - (\alpha+\beta)(M_A + M_B)] \\ &= \frac{N\mu^2}{3kT}[2B - (\alpha+\beta)M] \end{aligned} \tag{8.79}$$

This gives the susceptibility

$$\begin{aligned} \chi &= \frac{\mu_0 M}{B} \\ &= \frac{\dfrac{2\mu_0 N\mu^2}{3k}}{T + \dfrac{N\mu^2(\alpha+\beta)}{3k}} \\ &= \frac{C}{T+\theta} \end{aligned} \tag{8.80}$$

with

$$C = \frac{2\mu_0 N\mu^2}{3k}$$

$$\theta = \frac{N\mu^2(\alpha+\beta)}{3k} \qquad (8.81)$$

Equation (8.80) has the Curie–Weiss form, except that it has $T+\theta$ where the ferromagnet has $T-T_C$. The $1/\chi$ against T lines for the three types are compared in Fig. 8.10.

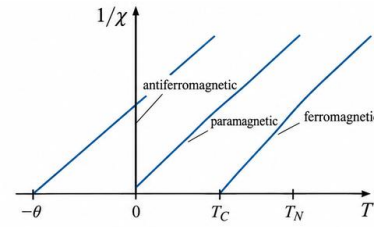


Fig. 8.10 (book): plot of 1/χ against T for para-, ferro- and antiferromagnetic substances above their critical temperatures.

The Néel temperature is not the same as θ:

$$\frac{T_N}{\theta} = \frac{\beta-\alpha}{\beta+\alpha} \qquad (8.82)$$

So T_N depends on both interactions. For $\alpha = 0$, $T_N = \theta$. For positive α, $T_N < \theta$, and T_N rises with β. Experiments confirm this.

Below T_N. Each sublattice is spontaneously magnetised, but the net magnetisation is zero. The susceptibility perpendicular to the easy direction, $\chi_\perp$, is almost constant. The susceptibility parallel to it, $\chi_\parallel$, falls to zero as $T \to 0$: at 0 K the internal field stops the spins from turning. At T_N, $\chi_\perp = \chi_\parallel = \chi$ (Fig. 8.11).

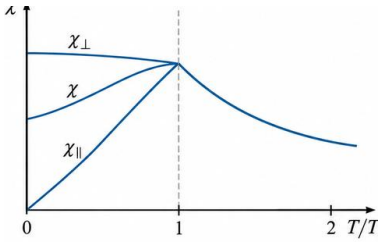


Fig. 8.11 (book): plot of χ against T/T_N for an antiferromagnetic single crystal.

3. Ferrimagnetism (§8.7)

A ferrimagnet is an antiferromagnet whose sublattice moments differ in size, so there is a net magnetisation. The typical material is a ferrite, an oxide of metals. The example is magnetite, Fe_3O_4 (or $FeO \cdot Fe_2O_3$). One Fe^{3+} ion occupies tetrahedral (A) sites, and the other Fe^{3+} and the Fe^{2+} occupy octahedral (B) sites. The A and B sublattices have opposite magnetisations, and the net moment of one formula unit equals that of one Fe^{2+} ion (Fig. 8.12). A unit cell of the spinel structure has eight formula units.

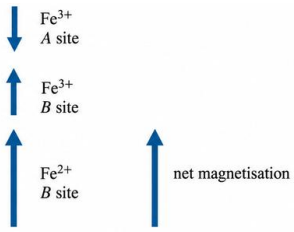


Fig. 8.12 (book): magnetisations of the individual sites and the net magnetisation in one formula unit of Fe_3O_4.

A ferrimagnet resembles a ferromagnet: both have spontaneous magnetisation, and both show hysteresis. Néel attributed the antiparallel arrangement to the negative AB interaction, with weaker AA and BB interactions. Above T_C both sublattice magnetisations follow the Curie–Weiss law; below it the saturation effects of Eq. (8.46) apply. The detailed theory is beyond the book's scope.

9.1–9.3 Polarisation, local field and dielectric constant (student notes)

Source: book-sources/bs-7-dielectrics.html, Puri and Babbar pp. 265–267. Equations 9.1 to 9.13 are cited by their book numbers. Figure 9.1 is redrawn.

Before you start

- A dielectric between capacitor plates and the dielectric constant $K = C/C_0$ (Plus Two recall below).
- Electric dipole moment $p = qd$ and the field of a dipole.
- Algebra of solving for one quantity in a ratio (Module IV pre-math, M2).

PLUS TWO RECALL: A DIELECTRIC IN A CAPACITOR

Recall. Putting a dielectric of constant K between the plates of a capacitor multiplies its capacitance by K: $C = KC_0$.

Re-derive.

1. The dielectric's dipoles are polarised and produce bound charges on its faces.
2. These bound charges oppose the field of the free charges, so the net field inside is smaller.
3. For the same charge, a smaller field means a larger voltage ratio, so C rises by K.

Saved. The reduction of the field by polarisation is Eq. (9.1), and the constant ε_r is Eq. (9.5).

1. Polarisation and susceptibility (§9.1)

A dielectric is an insulator with no free carriers. In an external field $\mathbf{E}_o$ its charges are displaced by distances smaller than an atomic diameter, so it forms a large number of dipoles aligned with the field. The material is polarised, with polarisation $\mathbf{P}$, the dipole moment per unit volume. The polarisation field $\mathbf{E}_p$ opposes the applied field, so the resultant field is smaller:

$$\mathbf{E} = \mathbf{E}_o + \mathbf{E}_p \tag{9.1}$$

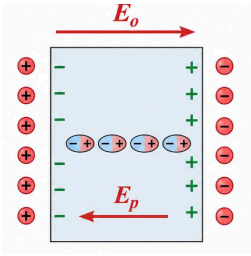


Fig. 9.1 (book): a dielectric slab in an external field E_o produced by fixed charges outside the slab; the internal polarisation field E_p is due to fictitious bound charges at the surface.

The field $\mathbf{E}_p$ is due to fictitious bound charges on the surfaces of the slab. For ordinary fields $\mathbf{P}$ is proportional to the macroscopic field $\mathbf{E}$, and in SI units

$$\mathbf{P} = \varepsilon_o \chi_e \mathbf{E} \tag{9.2}$$

where ε_o is the permittivity of free space and χ_e is the electric susceptibility. Except for a constant, χ_e measures the polarisation per unit resultant field.

2. The local field (§9.2)

The field acting on an atom is not the macroscopic field $\mathbf{E}$. It is the local field. For an atomic site of cubic symmetry, the Lorentz relation gives

$$\begin{aligned} \mathbf{E}_{loc} &= \mathbf{E}_o + \mathbf{E}_p + \frac{\mathbf{P}}{3\varepsilon_o} \\ &= \mathbf{E} + \frac{\mathbf{P}}{3\varepsilon_o} \end{aligned} \tag{9.3}$$

The local field includes the field of the polarisation of the other atoms, which is the extra term $\mathbf{P}/3\varepsilon_o$. The expression changes with the shape of the specimen.

3. Dielectric constant and polarisability (§9.3)

The electric displacement for an isotropic or cubic medium is

$$\begin{aligned} \mathbf{D} &= \varepsilon_o \varepsilon_r \mathbf{E} \\ &= \varepsilon_o \mathbf{E} + \mathbf{P} \end{aligned} \tag{9.4}$$

ε_r is the relative permittivity, or dielectric constant. It is dimensionless and, for an isotropic medium, a scalar. From (9.4):

$$\begin{aligned} \varepsilon_r &= \frac{\varepsilon_o \mathbf{E} + \mathbf{P}}{\varepsilon_o \mathbf{E}} \\ &= 1 + \chi_e \end{aligned} \tag{9.5}$$

So the dielectric constant measures the polarisation per unit resultant field, just as χ_e does.

The polarisability α of an atom is its dipole moment per unit local field:

$$p = \alpha E_{loc} \tag{9.6}$$

Polarisability is an atomic property. The dielectric constant is a macroscopic property that depends on how the atoms are arranged. For N atoms of equal α per unit volume,

$$P = Np = N\alpha E_{loc} \tag{9.7}$$

and in general

$$P = \sum_j N_j \alpha_j E_{loc}(j) \tag{9.8}$$

For cubic symmetry, (9.3) and (9.7) give

$$P = N\alpha \left(E + \frac{P}{3\varepsilon_o} \right) \tag{9.9}$$

Solving for P gives

$$\begin{aligned} \chi_e &= \frac{P}{\varepsilon_o E} \\ &= \frac{N\alpha/\varepsilon_o}{1 - \dfrac{N\alpha}{3\varepsilon_o}} \end{aligned} \tag{9.10}$$

and, using (9.5),

$$\varepsilon_r = 1 + \frac{N\alpha/\varepsilon_o}{1 - \dfrac{N\alpha}{3\varepsilon_o}}$$

This can be written as

$$\varepsilon_r = \frac{1 + \dfrac{2N\alpha}{3\varepsilon_o}}{1 - \dfrac{N\alpha}{3\varepsilon_o}} \tag{9.11}$$

which rearranges to the Clausius–Mossotti relation:

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N\alpha}{3\varepsilon_o} \tag{9.12}$$

In general form, for several kinds of atom,

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{1}{3\varepsilon_o} \sum_j N_j \alpha_j \tag{9.13}$$

This relates the macroscopic dielectric constant to the atomic polarisability, for cubic symmetry.

Section 7: sources of polarisability and frequency dependence (9.4–9.5, Example 9.1)

9.4–9.5 Sources of polarisability and its frequency dependence (student notes)

Source: book-sources/bs-7-dielectrics.html, Puri and Babbar pp. 267–272 (§9.4, §9.4.1 to §9.4.3, §9.5; Example 9.1). Equations 9.14 to 9.32 and 9.40 to 9.42 are cited by their book numbers. Figures 9.2 to 9.4 are redrawn.

Before you start

- The Clausius–Mossotti relation (Eq. 9.12) from §9.3.
- Potential energy of a dipole in a field, $U = -\mathbf{p}\cdot\mathbf{E}$ (Plus Two recall below).
- Simple harmonic motion: $\ddot{x} + \omega_0^2 x = 0$.

> **PLUS TWO RECALL: A DIPOLE IN AN ELECTRIC FIELD**
>
> **Recall.** An electric dipole of moment $\mathbf{p}$ in a uniform field $\mathbf{E}$ has potential energy $U = -\mathbf{p}\cdot\mathbf{E}$.
>
> **Re-derive.**
>
> 1. At angle θ, $U = -pE\cos\theta$, lowest when the dipole lies along the field.
> 2. Thermal motion spreads the orientations, so only a partial alignment remains.
> 3. The average alignment depends on pE/kT, which gives the orientational polarisability.
>
> **Saved.** Eq. (9.24) and the result (9.31).

1. Three sources of polarisability (§9.4)

The polarisability of a dielectric comes from three sources: electronic, ionic and dipolar (orientational). Which one matters depends on the material and on the frequency of the field.

2. Electronic polarisability (§9.4.1)

Electronic polarisation is the shift of an atom's electron cloud relative to its nucleus (Fig. 9.2). At optical frequencies the dielectric constant is due mainly to this. Replacing ε_r by n^2, where n is the refractive index, (9.13) becomes

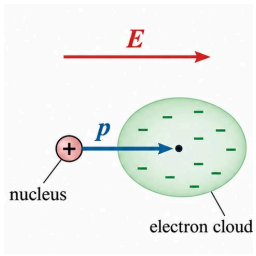


Fig. 9.2 (book): electronic polarisation in the presence of an external field.

$$\frac{n^2-1}{n^2+2} = \frac{1}{3\varepsilon_o}\sum_j N_j\alpha_j \quad \text{(electronic)} \tag{9.14}$$

Classical theory. An electron bound harmonically to an atom resonates at

$$\omega_o = \sqrt{\beta/m} \tag{9.15}$$

where β is the force constant. If the displacement under the local field is x:

$$-eE_{loc} = \beta x = m\omega_o^2 x \tag{9.16}$$

so the static electronic polarisability is

$$\alpha_e = \frac{p}{E_{loc}} = -\frac{ex}{E_{loc}} = \frac{e^2}{m\omega_o^2} \tag{9.17}$$

For a field $E_{loc}\sin\omega t$ at frequency ω, the equation of motion is

$$m\frac{d^2x}{dt^2} + m\omega_o^2 x = -eE_{loc}\sin\omega t \tag{9.18}$$

With $x = x_o\sin\omega t$:

$$m(-\omega^2 + \omega_o^2)x_o = -eE_{loc} \tag{9.19}$$

The dipole amplitude $p_o = -ex_o = e^2E_{loc}/[m(\omega_o^2 - \omega^2)]$ gives the frequency-dependent polarisability

$$\alpha_e = \frac{p_o}{E_{loc}} = \frac{e^2/m}{\omega_o^2 - \omega^2} \tag{9.20}$$

Quantum theory gives the same form, summed over transitions, with the oscillator strength f_{ij}:

$$\alpha_e = \frac{e^2}{m}\sum_j \frac{f_{ij}}{\omega_{ij}^2 - \omega^2} \tag{9.21}$$

3. Ionic polarisability (§9.4.2)

Ionic polarisation is the displacement of a charged ion relative to the others in a solid. For small displacements the force is simple harmonic, so

$$\beta \Delta x \cong eE \tag{9.22}$$

$$\alpha_i = \frac{p}{E} = \frac{e\Delta x}{E} \cong \frac{e^2}{\beta} \tag{9.23}$$

For $\beta \simeq 20$ N/m, $\alpha_i \simeq 10^{-39}$ F m^2. Ionic polarisation matters at low frequencies. Sodium chloride has $\varepsilon_r \simeq 5.6$ at low frequencies, falling to about 2.25 at optical frequencies.

4. Dipolar (orientational) polarisability (§9.4.3)

A polar molecule such as $\mathrm{H_2O}$ has a permanent dipole. With no field the orientations are random. A field orients the dipoles, and thermal agitation opposes this. In equilibrium the dipoles are spread over all angles. The energy at angle θ (Fig. 9.3) is

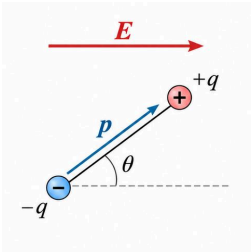


Fig. 9.3 (book): a molecule of dipole moment p placed in an electric field E.

$$U = -\mathbf{p}\cdot\mathbf{E} = -pE\cos\theta \tag{9.24}$$

The number of dipoles in the solid angle $d\Omega = 2\pi \sin\theta\, d\theta$ is proportional to

$$\exp\left(-\frac{U}{kT}\right) d\Omega = \exp\left(\frac{pE\cos\theta}{kT}\right) 2\pi\sin\theta\, d\theta \tag{9.25}$$

Each dipole contributes $p\cos\theta$ along the field. So the contribution to the polarisation is

$$2\pi p \exp\left(\frac{pE\cos\theta}{kT}\right)\cos\theta\sin\theta\, d\theta \tag{9.26}$$

and the average dipole moment per molecule is

$$\bar{p} = \frac{P}{N} = \frac{p\int_0^{\pi} \exp\left(\frac{pE\cos\theta}{kT}\right)\cos\theta\sin\theta\, d\theta}{\int_0^{\pi}\exp\left(\frac{pE\cos\theta}{kT}\right)\sin\theta\, d\theta} \tag{9.27}$$

With $x = pE/kT$ and $y = \cos\theta$, the same algebra as in §8.4.1 gives

$$\frac{\bar{p}}{p} = \coth x - \frac{1}{x} = L(x) \tag{9.28}$$

For $pE \ll kT$, $L(x) \simeq x/3$:

$$L(x) \simeq \frac{x}{3} = \frac{pE}{3kT} \tag{9.29}$$

So

$$P = N\bar{p} = \frac{Np^2E}{3kT} \tag{9.30}$$

and the dipolar polarisability per molecule is

$$\alpha_d = \frac{\bar{p}}{E} = \frac{p^2}{3kT} \tag{9.31}$$

Only molecules with a permanent dipole have this term. It falls with temperature. At room temperature $\alpha_d \sim 10^{-39}$ F m^2, which is comparable to the electronic value.

5. Frequency dependence of the total polarisability (§9.5)

The total polarisability is the sum of the three contributions:

$$\alpha = \alpha_e + \alpha_i + \alpha_d \tag{9.32}$$

It falls as the frequency rises (Fig. 9.4). Each process has a relaxation time, and it stops contributing once the frequency is high compared with the inverse of its relaxation time. Dipolar relaxation is slowest, so the dipolar part disappears first, then the ionic part, and the electronic part last.

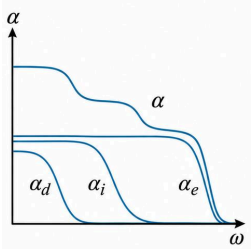


Fig. 9.4 (book): frequency dependence of the various contributions to polarisability.

6. Worked Example 9.1: ionic share of polarisability in NaCl

For NaCl the refractive index is $n = 1.5$ and the static dielectric constant is $\varepsilon_r = 5.6$. From the Clausius–Mossotti relation, with the static dielectric constant and the total polarisability $\alpha_e + \alpha_i$:

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N(\alpha_e + \alpha_i)}{3\varepsilon_0} \tag{9.40}$$

At optical frequencies, only the electronic part remains:

$$\frac{n^2-1}{n^2+2}=\frac{N\alpha_e}{3\varepsilon_0} \tag{9.41}$$

Dividing (9.41) by (9.40) gives

$$\left(\frac{n^2-1}{n^2+2}\right)\left(\frac{\varepsilon_r+2}{\varepsilon_r-1}\right)=\frac{\alpha_e}{\alpha_e+\alpha_i} \tag{9.42}$$

The percentage of ionic polarisability is

$$\begin{aligned}\left(\frac{\alpha_i}{\alpha_e+\alpha_i}\right)\times 100 &= \left[1-\frac{\alpha_e}{\alpha_e+\alpha_i}\right]\times 100\\ &= \left[1-\left\{\frac{(1.5)^2-1}{(1.5)^2+2}\right\}\left\{\frac{5.6+2}{5.6-1}\right\}\right]\times 100\\ &= 51.4\%\end{aligned}$$

Section 8: ferroelectricity and piezoelectricity (9.6–9.7)

9.6–9.7 Ferroelectricity and piezoelectricity (student notes)

Source: book-sources/bs-7-dielectrics.html, Puri and Babbar pp. 272–276 (§9.6 and §9.7). Equations 9.33 to 9.39 are cited by their book numbers. Figures 9.5 and 9.6 are redrawn. Item 22 is qualitative only; the algebra below is the book's picture of the transition, given for understanding.

Before you start

- Dielectric constant and the Clausius–Mossotti relation (§9.3, Eqs. 9.12, 9.13).
- Polarisability sources (§9.4).
- Ferromagnetism as the analogy: spontaneous magnetisation below T_C (§8.5).

Ferroelectricity and piezoelectricity are Beyond the Plus Two base (pre item P12). The Plus Two base gives only the idea of polarisation; the statements here are qualitative.

1. Ferroelectricity (§9.6)

Ferroelectricity is spontaneous polarisation, which means polarisation with no applied field. It is the electrical analogue of ferromagnetism. In a ferroelectric crystal the centres of positive and negative charge do not coincide even with no field, so there is a permanent dipole. Polarisation against field is not linear, and gives a hysteresis loop. Above the transition temperature, the Curie point T_C, the crystal becomes paraelectric and its dielectric constant falls sharply.

Barium titanate ($BaTiO_3$) and Rochelle salt ($NaKC_4H_4O_6\cdot 4H_2O$) are the familiar examples. The ferroelectrics fall into two groups. In the order–disorder group, the transition is the ordering of ions, and in hydrogen-bonded crystals the motion of protons matters (KH_2PO_4, RbH_2PO_4). In the displacive group, the transition is the displacement of one sublattice of ions relative to another ($BaTiO_3$, $KNbO_3$, $LiTaO_3$).

$BaTiO_3$. Above the Curie point (about 380 K) the crystal is cubic: Ba^{2+} at the corners, O^{2-} at the face centres and Ti^{4+} at the body centre (Fig. 9.5). The centres of charge coincide and there is no dipole. Below T_C, the Ti and Ba ions move up and the O ions move down, and the structure becomes tetragonal. The centres of charge then separate, which gives the spontaneous polarisation.

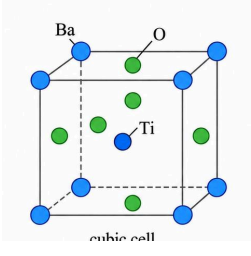


Fig. 9.5 (book): structure of $BaTiO_3$ for $T > T_C$.

In $\mathrm{BaTiO_3}$ the ferroelectricity comes from induced electronic and ionic dipoles. Rewriting (9.13) for this case gives

$$\varepsilon_r = \frac{1 + \dfrac{2}{3\varepsilon_o}\sum_j N_j\alpha_j}{1 - \dfrac{1}{3\varepsilon_o}\sum_j N_j\alpha_j} \tag{9.33}$$

where α_j is the electronic plus ionic polarisability of the jth ion. This assumes the local field at every atom is $\mathbf{E} + \mathbf{P}/3\varepsilon_o$. From (9.33), ε_r becomes infinite when

$$\sum_j N_j\alpha_j = 3\varepsilon_o \tag{9.34}$$

which allows a finite polarisation with no applied field. This is the condition for the *polarisation catastrophe*: the local field from the polarisation grows faster than the elastic restoring force on an ion. The shift is limited to a finite displacement by higher-order restoring forces.

Because ε_r is very sensitive to small departures from the critical value, put

$$\frac{1}{3\varepsilon_o}\sum_j N_j\alpha_j = 1 - 3s \tag{9.35}$$

with $s \ll 1$. Then (9.33) gives

$$\varepsilon_r = \frac{1}{s} - 2 \cong \frac{1}{s} \tag{9.36}$$

Near the transition take s to vary linearly with temperature, as might happen through thermal expansion of the lattice:

$$s \cong \frac{T - T_C}{\xi} \tag{9.37}$$

where ξ is a constant. Then

$$\varepsilon_r \cong \frac{\xi}{T - T_C} \tag{9.38}$$

This temperature dependence is close to the measured behaviour in the paraelectric state, as the Curie–Weiss law (8.64) is for ferromagnets.

2. Piezoelectricity (§9.7)

In some crystals an applied stress produces a net dipole moment, so polarisation charges appear on the surfaces. These are piezoelectric crystals, such as quartz, Rochelle salt and tourmaline. The inverse effect also occurs: an applied electric field strains the crystal.

In one-dimensional notation the piezoelectric equations are

$$\begin{aligned} P &= Zd + \varepsilon_o E\chi \\ e &= Zs + Ed \end{aligned} \tag{9.39}$$

where P is the polarisation, Z the stress, d the piezoelectric strain constant, E the field, χ the susceptibility, e the strain and s the compliance. The first gives polarisation from stress, and the second strain from field. The strains are small: in quartz a field of about 10^4 V/m gives a strain of about one part in 10^8.

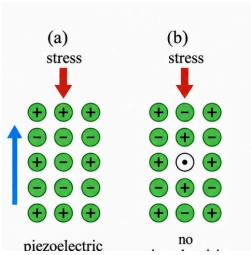


Fig. 9.6 (book): effect of crystal symmetry on piezoelectricity: (a) quartz, with no centre of inversion, shows piezoelectricity; (b) a crystal with a centre of inversion shows none.

Piezoelectricity needs the ions to be displaced so that the charge distribution loses its symmetry. The prerequisite is the absence of a centre of inversion (Fig. 9.6). Quartz has no centre of inversion and is piezoelectric. Every ferroelectric is piezoelectric, but not every piezoelectric is ferroelectric: $\mathrm{BaTiO_3}$ is both, quartz only piezoelectric.

Piezoelectric materials convert electrical energy into mechanical energy and back. They are used in gramophone pick-ups, microphones, strain gauges and ultrasonic generators.